Filing Fact-Checker

Project demonstration and technical walkthrough

Checking CrowdStrike statements against three official filings
with exact evidence, four verdicts and measured usage

Python 3.11+ / LangGraph / FastAPI / Streamlit
SQLite FTS5 / Local Ollama with Qwen 3.5 9B

Recorded evaluation: 20 statements × 3 configurations = 60 complete, correct runs

Task and output contract

A reviewer submits a statement. The system returns a verifiable report grounded only in the supplied filings.

Verdict	Decision rule	Example
SUPPORTED	Evidence confirms the same fact, scope and period	7,925 full-time employees on 31 Jan 2024
CONTRADICTED	Evidence refutes the same fact, scope and period	9,000 employees on that same date
OUTDATED	An earlier filing supports a current claim, but a later filing changes it	Latest cash stated as the July balance
NOT_ENOUGH_INFO	Available disclosures do not resolve the claim	Exactly 5,000 US-based employees

Every report contains

Statement, verdict, explanation, exact quotations, filing IDs and PDF pages.
Execution status, stop reason, trace, research plan, model/tool calls, tokens and time.

Evidence corpus and financial context

The manifest identifies both the reporting period and the date the company filed each report.

Included report	Period ended	Filed on	Role in verification
10-K annual report	31 Jan 2024	7 Mar 2024	Business, workforce, annual financials and notes
10-Q quarterly report	31 Jul 2024	29 Aug 2024	Interim balances and developments
10-Q quarterly report	31 Oct 2024	27 Nov 2024	Latest supplied quarter and updated balances

Interpretation rules

Fiscal years end on January 31. Fiscal 2025 largely overlaps calendar 2024.
Financial tables often report USD thousands. A row value needs its scale and column heading.

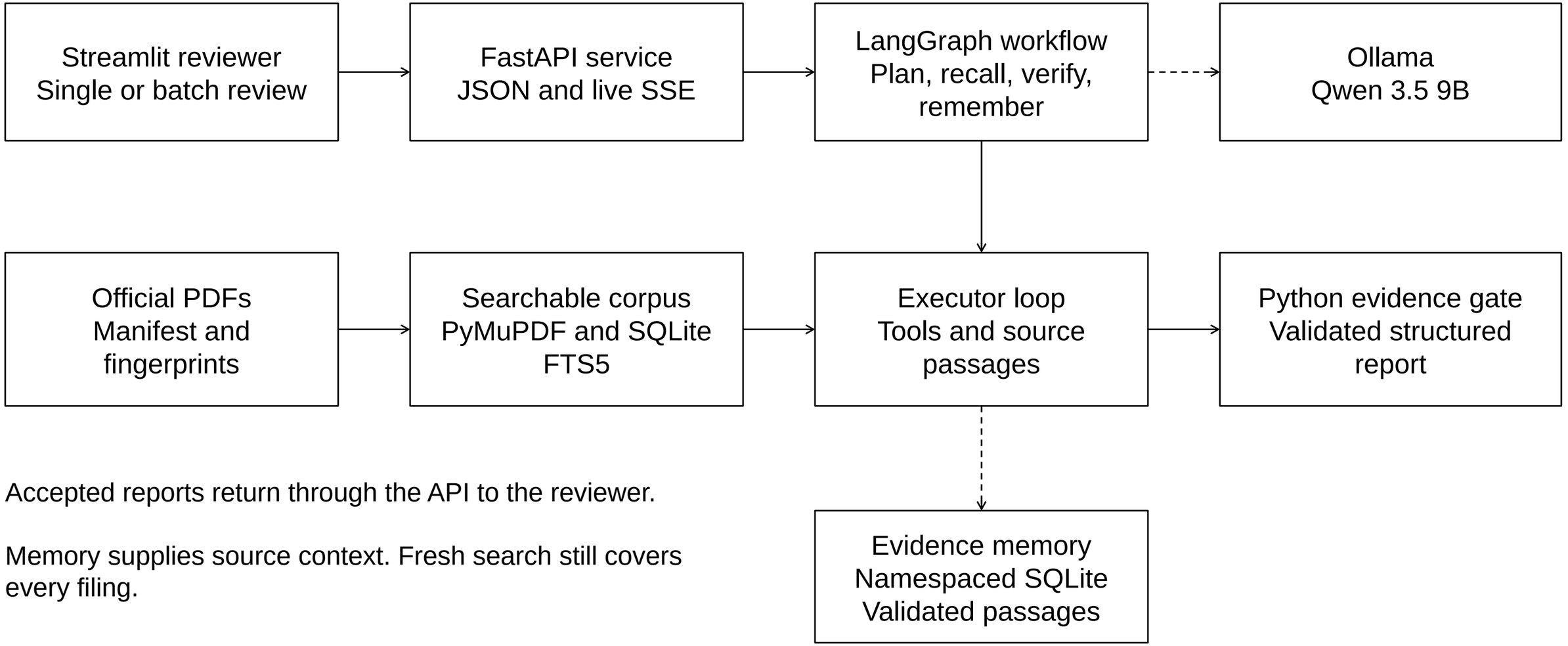
Evidence boundary

“Latest” means latest within these files.
PDF page numbers refer to the included copies. A quarterly report can repeat older comparative balances.

The filing date orders sources. The table date identifies the value. Both matter for temporal claims.

Complete system architecture

The reviewer uses HTTP. The CLI and evaluator invoke the same Python workflow directly.



Accepted reports return through the API to the reviewer.

Memory supplies source context. Fresh search still covers every filing.

PDF preparation and evidence retrieval

A local lexical index keeps the corpus searchable without an embedding service.

Preparation

1. Read manifest and fingerprint each PDF.
2. Extract page text with PyMuPDF.
3. Split into 1,800-character chunks with 300-character overlap.
4. Retain page context, offsets and stable passage IDs.
5. Reuse parsed caches and build SQLite FTS5 index.

Search and reading

The bootstrap combines the original statement with the first planning query. BM25 ranks matches.

Reciprocal rank fusion combines result lists using 1 / (60 + rank). Default retrieval loads one passage per filing. Further tool calls can fill specific gaps.

Canonical passage field	Purpose
10k_2024_01_31:p22:c1	Stable filing / PDF page / chunk identity
pdf_sha256 + source_url	Source authentication and original filing link
pdf_page + page_start / page_end	Page navigation and character provenance
page_context + is_toc	Heading/unit context and table-of-contents filtering

MERMAID adaptation and agent responsibilities

A compact claim plan guides a bounded Executor. Python owns deterministic operations.



Component	**Implemented responsibility**	**Why this choice fits the task**
Decomposer / LLM	Subject, relation, object, attribution, time, topics and queries	A typed plan preserves scope without a full knowledge graph
Executor / LLM + graph	Interpret evidence, choose tools, propose verdict and citations	Financial meaning still needs model judgment
Toolset / Python	Search, read passages, load page context, validate arguments	Registry makes tools discoverable and execution controlled
Memory / Python	Recall and persist authenticated source evidence	Related claims reuse passages without inheriting an answer

Simplifications: local filing tools replace web tools. Lexical recall replaces semantic reranking.
Planning remains optional for the single-agent comparison.

Tool selection and the execution loop

The model receives each tool’s name, description and argument schema before it chooses a call.

Tool or action	Inputs and work	When it helps
search_and_read	Query, filing selection and result limits. Search and return selected passages.	Find a missing metric or newer disclosure
read_passage	Canonical passage_id. Return its original text and metadata.	Inspect a known result in full
read_page_context	Filing and PDF page. Return surrounding page context.	Recover a table heading, unit or adjacent context
submit_verdict	Verdict, explanation and passage_id / quote pairs. Run evidence gate.	Finalize only when the candidate validates

One model turn

1. Read claim, plan, evidence and prior feedback.
2. Request a named tool with structured arguments.
3. Python validates and executes it.
4. Append the tool result before the next model request.

Clear stopping rules

Default: 8 model calls and 16 tool calls total.
The final model slot requests submission.
Two searches with no new passages trigger submission-only mode. Rejections use the remaining budget.

Working state and message accumulation

The next model request sees the accumulated observations and feedback from the current run.

Stage	Conversation/state update	What the next turn can use
Initial context	System rules + user statement + validated plan	Claim, corpus dates, scope and stopping rules
Bootstrap retrieval	Python adds selected source passages and IDs	Evidence from a search across all supplied filings
Assistant tool request	Structured name and arguments enter the message trail	What the model requested and why more context is needed
Tool observation	Append source text, IDs or error / rejection feedback	New facts and corrections to the attempted submission
Accepted candidate	Assemble typed result, trace and usage counters	Final JSON report and evidence for persistence

Temporary state

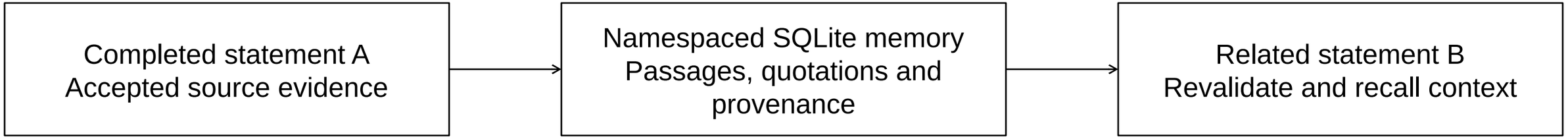
Messages, plan, read IDs, searched filings, usage, search progress and candidate result exist for one invocation.

Durable storage

SQLite corpus and evidence memory survive restarts. The graph has no persistent checkpoint or resumable conversation store.

Evidence memory and source freshness

Memory reuses validated passages across related statements while every run still searches all filings.



Recall controls

FTS stemming and topical terms select related evidence.
Exclude date-only, number-only and generic unit matches.
Require agreement on up to three topical terms.
Recall at most three passages and check current source fingerprints.

Freshness and isolation

Namespaces separate evaluation and reviewer sessions.
Memory stores source evidence rather than verdicts.
Fresh retrieval includes every supplied filing.
Newer source evidence can override an older current-state claim. Persist only after a complete result.

Measured full versus memory-off: 54 versus 58 model calls, 74 versus 78 tool calls, 191,196 versus 192,919 tokens. Gains depend on claim order and relevance.

Structured submission and evidence validation

A fluent answer becomes a completed report only after the Python evidence gate accepts it.

Gate	Check	Response to failure
Schema	Allowed verdict, explanation and structured citation pairs	Return validation details for a corrected submission
Read coverage	Every supplied filing searched and cited passage IDs loaded	Require missing search or read work
Quote provenance	Continuous source match with whitespace / quotation-style tolerance	Reject invented, stitched or changed quotations
Financial consistency	Metric, scale, decimal precision and comparative period columns	Return resolved row/column context where supported
Scope and chronology	Relevant population, reporting coverage, earlier/later evidence	Reject invalid scope or unsupported OUTDATED classification
Final consistency	Explanation agrees with verdict and evidence	Request a coherent result within the remaining budget

The report restores original source punctuation and attaches filing, PDF page, URL and fingerprint.
Citation authentication establishes provenance. Broader semantic entailment still needs model judgment.

Financial units, time and population scope

The same number can support a different metric, date or population. The system preserves those distinctions.

Check	Source context	Decision example
Scale and rounding	FY2024 total revenue: 3,055,555 in USD thousands	USD 3.055555 billion rounds to USD 3.06 billion
Metric identity	Subscription revenue: 2,870,557 in USD thousands	USD 3,055,555,000 describes total revenue, not subscription
Comparative columns	October cash row also repeats the January balance	Select the column matching the statement’s date
Historical versus latest	July cash: USD 4.038536B. October cash: USD 4.260324B	July-dated claim is SUPPORTED. Calling July “latest” is OUTDATED
Population scope	7,925 full-time employees is the company total	It does not establish exactly 5,000 US employees
Reporting coverage	Latest supplied filing predates a later completed annual period	Interim actuals do not refute undisclosed full-year actuals

Decimal arithmetic handles supported scales and precision intervals. Inequalities, negations and broader financial interpretation remain model-dependent.

FastAPI contracts and streaming events

POST /verify returns JSON. POST /verify/stream streams steps and the final result through SSE.

Request body

```
{
  "statement": "CrowdStrike’s fiscal year ends on January 31.",
  "use_decomposer": true,
  "use_memory": true,
  "memory_namespace": "presentation",
  "max_model_calls": 8,
  "max_tool_calls": 16
}
```

Streaming lifecycle

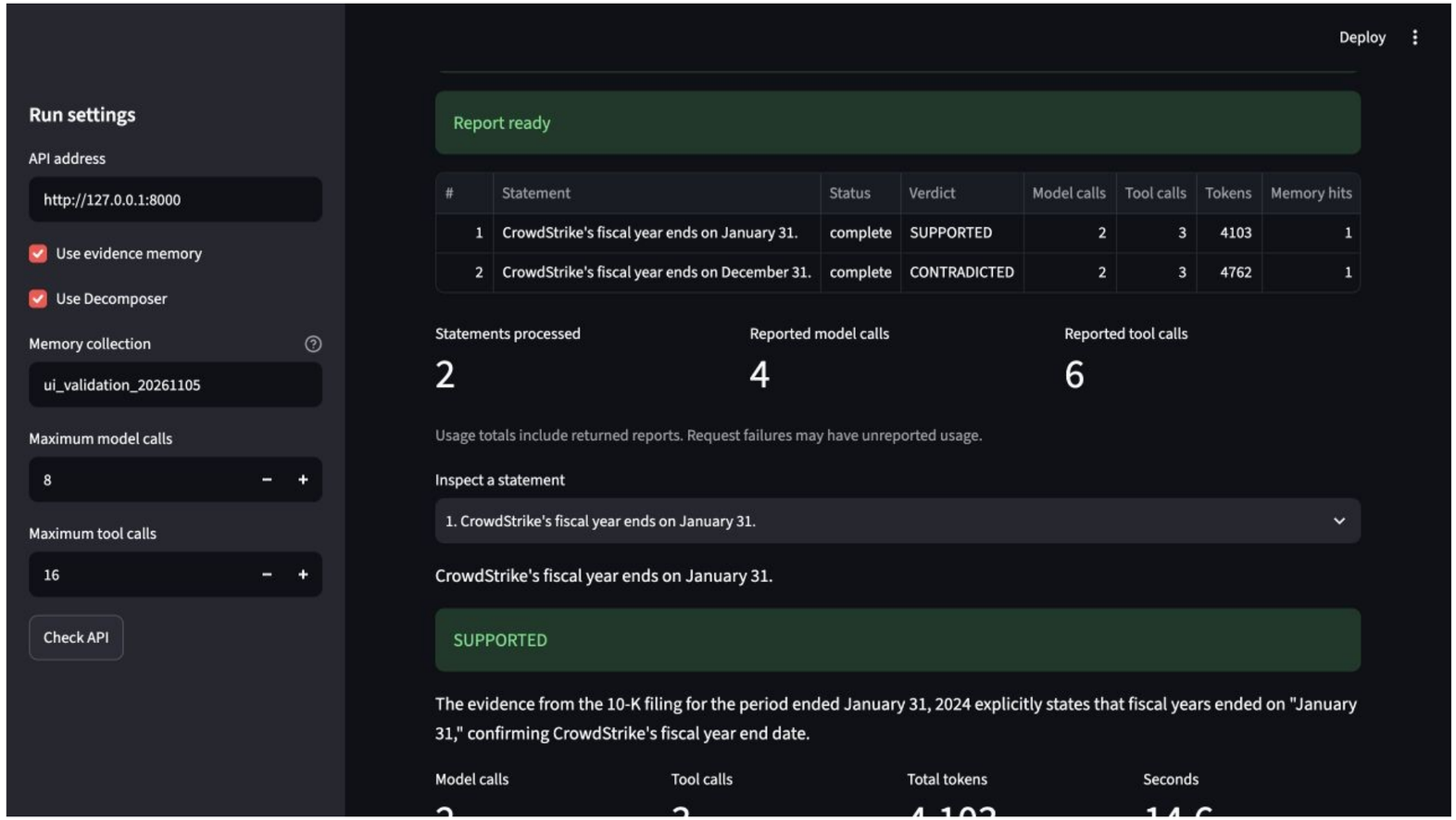
queued
started
step events as nodes and tools finish
result containing the validated report
done with outcome=result

On service failure: error, then done.
All events carry the same run_id.

Route / behavior	Purpose
GET /health and GET /docs	Service status and interactive Swagger UI
GET /openapi.json	Machine-readable API contract
Per-process asyncio lock	Serialize API verification runs for the local model
Input validation	Reject empty statements, invalid settings and incompatible budgets

Reviewer interface and demo controls

The actual Streamlit reviewer exposes results, evidence, live events and usage for one statement or a batch.



Recorded interface validation. The displayed counters belong to this captured run.

Demonstration sequence

1. Open localhost:8501. Check API.
2. Enable memory and Decomposer.
3. Submit a claim. Watch live steps.
4. Inspect evidence and PDF page.
5. Expand the JSON report.
6. Run a batch. Download its report.

Settings expose namespaces and call limits. Batch review sends sequential single-statement requests.

Recorded demo: supported and contradicted claims

Changing one value produces a different verdict against the same source and date.

SUPPORTED / T03

Claim: “CrowdStrike had 7,925 full-time employees as of January 31, 2024.”

The 10-K directly confirms the headcount, employment type and date.

CONTRADICTED / T04

Claim: “CrowdStrike had 9,000 full-time employees as of January 31, 2024.”

The same disclosure gives 7,925, which refutes 9,000 for that scope.

Source quotation

“As of January 31, 2024, we had 7,925 full-time employees.”

10-K for period ended 31 Jan 2024 / PDF page 22 / 10k_2024_01_31:p22:c1

Recorded full workflow	Model calls	Tool calls	Tokens	Seconds
T03 / complete / SUPPORTED	2	3	4,507	17.811
T04 / complete / CONTRADICTED	2	3	4,945	21.955

Recorded demo: a balance becomes outdated

T15: “CrowdStrike’s most recently reported cash and cash equivalents are USD 4,038,536,000.”

Evidence	July report	October report
Period ended	31 Jul 2024	31 Oct 2024
Source and PDF page	10q_2024_07_31 / page 6	10q_2024_10_31 / page 9
Cash row, in USD thousands	4,038,536	4,260,324
Normalized amount	USD 4,038,536,000	USD 4,260,324,000
Role	Supports the earlier amount	Updates the same metric in a later filing

Accepted verdict: OUTDATED

The claim calls the July amount “most recently reported.” The October filing updates cash to USD 4,260,324,000. The final report cites both sources.

A claim explicitly dated 31 July 2024 remains historically supported.
Recorded T15: complete, 2 model calls, 3 tool calls, 5,899 tokens, 48.495 seconds.

Recorded demo: missing information and execution status

T18: “CrowdStrike had exactly 5,000 US-based full-time employees as of January 31, 2024.”

Available evidence

The 10-K reports 7,925 full-time employees for the company on that date. The cited disclosure does not establish the exact US-based subset.

Result: NOT_ENOUGH_INFO, status=complete.

Why the distinction matters

A company-wide total can exceed a regional count. The global figure alone neither confirms nor refutes exactly 5,000 US employees.

The quotation gives related context. It does not prove corpus-wide absence.

Result condition	Meaning	Evaluation treatment
complete + NOT_ENOUGH_INFO	Search and reasoning finish without resolving the claim	Correct only when the expected label matches
incomplete + fallback label	Model failure, step limit or repeated malformed output	Always incorrect, even if fallback label matches

Recorded T18: 3 model calls, 4 tool calls, 10,443 tokens, 51.127 seconds.
Other missing-information cases cover attrition and an unsupported fiscal 2030 forecast.

Failure handling and bounded execution

Errors become explicit feedback or an incomplete outcome with a stop reason.

Failure or edge case	Implemented behavior
Malformed Decomposer plan	Validate structured fields. Fall back to the original statement.
Bad tool arguments or unavailable passage	Return an error observation and allow a corrected call within budget.
Unsupported or altered quotation	Reject the candidate and preserve original source punctuation.
Repeated searches with no new evidence	Request a final submission rather than continuing retrieval indefinitely.
Model failure, repeated unstructured output, exhausted limits	Return incomplete with a specific stop reason. Count unsuccessful attempts.
PDF table headers, repeated text and TOC entries	Carry page context and provenance. Filter TOC results. Allow page-context reads.
Source changes or stale memory	Use corpus fingerprints and revalidate recalled passages before reuse.

Known boundaries: lexical retrieval, text PDFs, selected numeric formats and model interpretation.
OCR, semantic reranking and independent compound-claim verification remain extensions.

Evaluation protocol and complete benchmark

20 unique development statements, each evaluated under three configurations, produce 60 verification runs.

Configuration	Decomposer	Memory	Correct / complete	Model calls	Tool calls	Tokens	Seconds
Full	On	On	20 / 20	54	74	191,196	839.5
No memory	On	Off	20 / 20	58	78	192,919	963.1
Single agent	Off	On	20 / 20	35	54	158,559	772.2

Dataset coverage

7 supported, 7 contradicted, 3 outdated and 3 missing-information statements.
Fiscal dates, workforce, revenue, concentration, incident disclosures, cash, assets and liabilities.

Comparison controls

Same ordered claims and total call budgets.
Each configuration starts with an isolated empty memory namespace. Labels and reference answers stay in the evaluation client.

Accuracy requires both a matching verdict and complete status. Saved predictions include exact evidence, explanations and usage. This is a development benchmark used during iteration, not a held-out test.

Measured efficiency and ablation findings

The saved original baseline remains available for comparison with the final implementation.

Full-workflow metric	Original baseline	Final run	Change
Accuracy / completed	85% / 18 of 20	100% / 20 of 20	All cases complete and correct
Model calls	61	54	11.5% fewer
Tool calls	81	74	8.6% fewer
Total tokens	266,260	191,196	28.2% fewer
Elapsed seconds	1,017.1	839.5	17.5% lower

Memory comparison

Full versus memory-off saves 4 model calls and 4 tool calls. Tokens fall 0.9% and elapsed time falls 12.8% in the recorded run.
Relevance filters and deduplicated source text reduce context waste.

Planning comparison

Single-agent mode uses 35 model calls versus 54, with the same development accuracy. Planning adds measurable overhead.
Its elapsed time is 8.9% above its own original baseline despite using fewer tokens.

Measurements: Apple M5, 24 GiB RAM, warm local model, one recorded run. Latency varies with hardware and load.

Run instructions, validation and project handoff

The repository contains the application, source PDFs, labelled claims, predictions, architecture and this demonstration.

Local demonstration setup

1. Install Python dependencies and pull qwen3.5:9b.
2. Prepare the corpus: python -m filing_checker.prepare
3. Start API in one terminal:
python -m uvicorn filing_checker.api:app --host 127.0.0.1 --port 8000
4. Start UI in another terminal:
python -m streamlit run streamlit_app.py
5. Open localhost:8501 and 127.0.0.1:8000/docs.

Recorded verification

60 / 60 complete, correct benchmark runs
92 automated tests
12 / 12 additional live API checks
Swagger, streaming, single/batch review and JSON downloads exercised successfully.

Evaluation commands

python -m unittest discover -s tests -v
python -m filing_checker.evaluate --validate-only
python -m filing_checker.evaluate --mode all

Next extensions

Semantic reranking, OCR, independent compound claims and durable graph resume. The current system retains explicit evidence and execution limits.

Repository: github.com/cyash24f3/fillingfactz
Detailed Excalidraw board: docs/architecture/ Recorded outputs: results/benchmark/